MA3K6: Boolean Functions
Module Notes and Summary

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Contents

1 Lecture 1: Basic Properties of Boolean Functions

Definition 1.1.
1. A *Boolean function* (on n variables) is a function $f : \{0,1\}^n \to \{0,1\}$
2. The set of points in $\{0,1\}^n$ where the function f is 1 is called *True points* denoted $T(f)$.
3. The set of points in $\{0,1\}^n$ where the function f is 0 is called *False points* denoted $F(f)$.

Theorem 1.2. *There are 2^{2^n} n-variable Boolean functions.*

Definition 1.3. Given n variables $x_1, \dots, x_n$, a *Boolean expression* is defined recursively as follows:

1. Each literal 0, 1, $x_1, \dots, x_n$ is a Boolean expression.
2. If φ is a Boolean expression, then $\overline{\varphi}$ (negation) is also a Boolean expression.
3. If φ_1, φ_2 are Boolean expressions, then $\varphi_1 \vee \varphi_2$ and $\varphi_1 \wedge \varphi_2$ are Boolean expressions.

Every Boolean functions can be represented as a Boolean expression.

Example 1.4. Let $f(x, y, z) = (x = y) \vee (x \neq z)$. This is currently not a Boolean expression so we turn it into one. We have

$$(x = y) \vee (x \neq z) = ((x \wedge y) \vee (\overline{x} \wedge \overline{y})) \vee ((x \wedge \overline{z}) \vee (\overline{x} \wedge z))$$

and the LHS is indeed a Boolean expression.

Remark 1.5.
1. Boolean functions can be represented as Boolean expressions and truth tables with each Boolean point listed in lexicographical order.
2. Two Boolean expressions can represent the same Boolean function. An example being the De Morgan laws: $\overline{x \vee y} = \overline{x} \wedge \overline{y}$

2 Lecture 2: Equivalence, DNF, CNF, Duality

Definition 2.1. Two Boolean expressions, φ_1, φ_2, are *equivalent* if and only if $f_{\varphi_1} = f_{\varphi_2}$

Theorem 2.2. *Some basic identities are as follows:*

1. $x \vee (x \wedge y) = x = x \wedge (x \vee y)$
2. $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$ *and* $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$
3. $\overline{(x \vee y)} = \overline{x} \wedge \overline{y}$ *and* $\overline{(x \wedge y)} = \overline{x} \vee \overline{y}$ *(De Morgan Laws)*

Remark 2.3. We can represent Boolean expressions as a Venn-Diagram. For the expression $x \vee y$ we can say the X circle represents true and outside represents false. Same for the Y circle. Then the region $X \cup Y$ represents the truth values of this expression.

Definition 2.4. A *disjunctive normal form* (DNF) is a disjunction of *terms* where a *term* is a conjunction of literals.

Definition 2.5. A *conjunctive normal form* (CNF) is a conjunction of *clauses* where a *clause* is a disjunction of literals.

Definition 2.6. When every term/clause uses all n literals in the domain, the DNF/CNF is called *canonical.*

Canonical DNFs, CNFs are unique up to reordering.

Example 2.7. This example motivates the following proposition. Consider the following truth table:

x_1	x_2	x_3	f	Related Term	Related Clause
0	0	0	0	N/A	$x_1 \vee x_2 \vee x_3$
0	0	1	1	$\overline{x_1}\,\overline{x_2}x_3$	N/A
0	1	0	0	N/A	$x_1 \vee \overline{x_2} \vee x_3$
0	1	1	1	$\overline{x_1}x_2x_3$	N/A
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	1	1	1	$x_1x_2x_3$	N/A

Proposition 2.8. *Every Boolean function admits a Boolean expression φ such that $f = f_\varphi$*

Proof. Suppose f is an n-dimensional Boolean function. Consider the following DNF

$$\varphi(x_1, \ldots, x_n) = \bigvee_{Y \in T(f)} \left(\bigwedge_{i:x_i=1} x_i \wedge \bigwedge_{j:x_j=0} \overline{x_j} \right)$$

Then $f_\varphi = f$ since $T(f_\varphi) = T(f)$ by construction of φ as shown in the example above. Similarly, one may also represent f by the following CNF

$$\varphi(x_1, \ldots, x_n) = \bigwedge_{Y \in F(f)} \left(\bigvee_{i:x_i=0} x_i \vee \bigvee_{i:x_i=1} \overline{x_i} \right)$$

Then $f_\varphi = f$ since $F(f_\varphi = F(f)$ by construction of φ in this case. $\square$

Example 2.9. A function f with vector representation $\overline{f} = 01110010$ has canonical DNF as

$$(\overline{x_1}\,\overline{x_2}x_3) \vee (\overline{x_1}x_2\overline{x_3}) \vee (\overline{x_1}x_2x_3) \vee (x_1x_2\overline{x_3})$$

and has canonical CNF as

$$(x_1 \vee x_2 \vee x_3)(\overline{x_1} \vee x_2 \vee x_3)(\overline{x_1} \vee x_2 \vee \overline{x_3})(\overline{x_1x_2x_3})$$

Example 2.10. The following expression $x_1x_2 \vee x_1x_3$ has vector representation $f = 00000111$ which has canonical DNF

$$(x_1\overline{x_2}x_3) \vee (x_!x_2\overline{x_3}) \vee (x_1x_2x_3)$$

Definition 2.11. Given a Boolean function $f(x_1, \ldots, x_n)$ we define the *dual* of f as

$$f^d := \overline{f(\overline{x_1}, \ldots, \overline{x_n}}$$

Example 2.12. The OR and AND operators are duals of each other (using De-Morgan laws):

1. $(x \vee y)^d = \overline{\overline{x} \vee \overline{y}} = x \wedge y$
2. $(x \wedge y)^d = \overline{\overline{x} \wedge \overline{y}} = x \vee y$

Theorem 2.13. *If f and g are Boolean functions, then*

1. $(f^d)^d = f$
2. $(\overline{f})^d = \overline{f^d}$
3. $(f \vee g)^d = f^d \wedge g^d$
4. $(f \wedge g)^d = f^d \vee g^d$

3 Lecture 3: Duality, Orthogonal, Implicants

Definition 3.1. The *dual* of a Boolean expression φ, can be yielded by interchanging $\vee$ and $\wedge$ operators as well as 0 and 1 literals. It is denoted as φ^d.

The following lemma comes directly from the recursive definition of Boolean functions. (Try an example)

Lemma 3.2. *For all Boolean expressions, φ_1, φ_2, we have*

1. $f_{\varphi_1 \vee \varphi_2} = f_{\varphi_1} \vee f_{\varphi_2}$
2. $f_{\varphi_1 \wedge \varphi_2} = f_{\varphi_!} \wedge f_{\varphi_2}$

Theorem 3.3. *For every Boolean expression φ, we have $f_{\varphi^d} = (f_\varphi)^d$. (If φ represents f then φ^d represents f^d)*

Proof. We induct on $\#t$, the number of operations in φ.
Let $t = 0$, then WLOG, $\varphi = x$ or $\varphi = 0$. Hence $\varphi^d = x$ or $\varphi^d = 1$. So, $f_{\varphi^d} = x$ or $f_{\varphi^d} = 1$. And,

$$(f_\varphi)^d = \overline{\overline{x}} = x = f_{\varphi^d} \text{ or } (f_\varphi)^d = \overline{0} = 1 = f_{\varphi^d}$$

this proves the base case. If $t \geq 1$, then $\varphi = \varphi_1 \vee \varphi_2$ or $\varphi = \varphi_1 \wedge \varphi_2$ or $\varphi = \overline{\varphi_1}$.

1. If $\varphi = \varphi_1 \vee \varphi_2$, then

   $$\varphi^d = \varphi_1^d \wedge \varphi_2^d \implies f_{\varphi^d} = f_{\varphi_1^d \wedge \varphi_2^d} = f_{\varphi_1^d} \wedge f_{\varphi_2^d}$$

   Then applying our inductive hypothesis $f_{\varphi_1^d} = (f_{\varphi_1})^d$ and $f_{\varphi_2^d} = (f_{\varphi_2})^d$ so

   $$f_{\varphi_1^d} \wedge f_{\varphi_2^d} = (f_{\varphi_1})^d \wedge (f_{\varphi_2})^d$$

   so using (2.13), we have

   $$(f_{\varphi_1})^d \wedge (f_{\varphi_2})^d = (f_{\varphi_1} \vee f_{\varphi_2})^d = (f_{\varphi_1 \vee f_{\varphi_2}})^d$$

   as required.

2. The case for $\varphi = \varphi_1 \wedge \varphi_2$ is the same as the previous case.
3. If $\varphi = \overline{\varphi_1}$, then $\varphi^d = \overline{\varphi_1}^d = \varphi$ so $f_{\varphi^d} = f_\varphi = f_\varphi^d$

$\square$

Consider a DNF

$$\varphi = \bigvee_{k=1}^{m} C_k = \bigvee_{k=1}^{m} \left(\bigwedge_{i \in A_k} x_i \wedge \bigwedge_{j \in B_k} \overline{x_j} \right) \text{ where } A_k \cap B_k = \emptyset \text{ for all } k \tag{1}$$

Definition 3.4. A DNF of the form (1) is *orthogonal* (the sum of *disjoint products*) if every two distinct terms of the DNF are *conflicting* in at least one variable, so there must be a variable that appears complemented in one of the terms and uncomplemented in the other term. We can write that as $(A_k \cap B_\ell) \cup (A_\ell \cap B_k) \neq \emptyset$

Remark 3.5. 1. Two conflicting terms cannot be simultaneously true from this definition.

2. Hence, if φ is orthogonal, then $T(\varphi)$ is the **disjoint union** of the sets of true points of its terms.

Example 3.6. Any Canonical DNF is an orthogonal DNF

Corollary 3.7. *Since every Boolean function has a Boolean expression in the form of a canonical DNF, then every Boolean function admits an orthogonal Boolean expression.*

Theorem 3.8. *If φ is represented by an orthogonal DNF as shown in (1), then the number of true points of φ is*

$$|T(\varphi)| = \sum_{k=1}^{m} 2^{n-|A_k|-|B_k|}$$

Proof. By our remark, $T(\varphi) = \coprod_{k=1}^{m} T(C_k)$, we have $|T(f)| = \sum_{k=1}^{m} |T(C_k)|$. Now consider the following example that justifies $|T(C_k)|$:
Expression: $x_1\overline{x_3}x_7$, 7 variables so $n = 7$ so if $x_1 = 1, x_3 = 0, x_7 = 1$ then we have $7 - 3 = 7 - |A_i| - |B_i|$ free choices hence $2^{n-|A_i|-|B_i|} = |T(C_i)|$. So $|T(\varphi)| = \sum_{k=1}^{m} 2^{n-|A_k|-|B_k|}$ as required. $\square$

Definition 3.9. Given Boolean functions f, g on $\mathcal{B}^n := \{0,1\}^n$, we say f *implies* g if and only if

$$\forall\ X \in \mathcal{B}^n \quad \text{, we have } f(X) = 1 \implies g(X) = 1$$

or equivalently, $T(f) \subseteq T(g)$. This is denoted by $f \leq g$.

Theorem 3.10. *For all Boolean functions, f, g on $\mathcal{B}^n$, the following are equivalent*

1. $f \leq g$
2. $f \vee g = g$
3. $\overline{f} \vee g = \mathbf{1}_n$
4. $f \wedge g = f$
5. $f \wedge \overline{g} = \mathbf{0}_n$

Proof. We can simply check each one using a Venn Diagram. $\square$

Theorem 3.11. *For all Boolean functions f, g, h on $\mathcal{B}^n$, we have*

1. $\mathbf{0}_n \leq f \leq \mathbf{1}_n$
2. $f \wedge g \leq f \leq f \vee g$
3. $f = g$ *if and only if* $f \leq g$ *and* $g \leq f$ *so* $T(f) = T(g)$ *hence* $F(f) = F(g)$
4. $(f \leq h$ *and* $g \leq h$ *if and only if* $f \vee g \leq h$
5. $(f \leq g$ *and* $f \leq h)$ *if and only if* $f \leq g \wedge h$
6. *if* $f \leq g$ *then* $f \wedge h \leq g \wedge h$
7. *if* $f \leq g$ *then* $f \vee h \leq g \vee h$

4 Lecture 4: Implicants, Prime Implicants

Theorem 4.1. *Consider two terms given by*

$$C_{AB} = \left(\bigwedge_{i \in A} x_i\right) \wedge \left(\bigwedge_{j \in B} \overline{x_j}\right) \text{ and } C_{FG} = \left(\bigwedge_{i \in F} x_i\right) \wedge \left(\bigwedge_{j \in G} \overline{x_j}\right) \text{ with } A \cap B = \emptyset,\ F \cap G = \emptyset$$

Then $C_{AB} \leq C_{FG}$ if and only if $F \subseteq A$, $G \subseteq B$

Proof. Suppose $F \subseteq A$, $G \subseteq B$ and let $Y = (y_1, \ldots, y_n) \in T(C_{AB})$. Then $y_i = 1$ for all $i \in A$ and $y_j = 0$ for all $j \in B$. Therefore, $y_i = 1$ for all $i \in F \subseteq A$ and $y_j = 0$ for all $j \in G \subseteq B$. Hence $Y \in T(C_{FG})$ which means $C_{AB} \leq C_{FG}$.
Conversely, suppose $C_{AB} \leq C_{FG}$ so that $T(C_{AB}) \subseteq T(C_{FG})$ and assume, for a contradiction, that $F \not\subseteq A$. Set $y_i = 1$ for all $i \in A$ and $y_j = 0$ for all $j \notin A$. Then $Y = (y_1, \ldots, y_n) \in T(C_{AB})$ since $y_j = 0$ for all $j \in B$ $(A \cap B = \emptyset)$. But we claim $Y \notin T(C_{FG})$ since $F \not\subseteq A$ so $\exists\ y^* \in F \backslash A$ with $y^* = 0$ and therefore, $\bigwedge_{i \in F} x_i = 0$ hence $C_{FG}(Y) = 0$ which is a contradiction. Similarly, $G \subseteq B$ thus we are done. $\square$

Example 4.2. $xy\overline{z}\overline{w} \leq x\overline{z}$ by the above theorem.

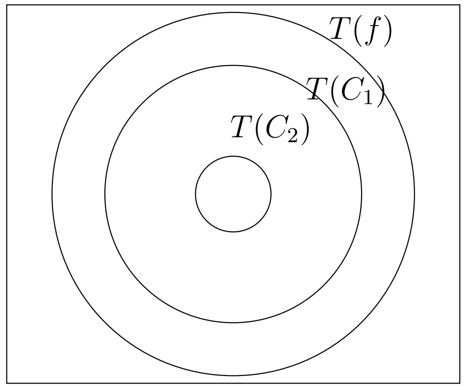
Definition 4.3. Let f be a Boolean function and C a term. We say C is an *implicant* of f if $f_C \implies f$ ($C \leq \varphi$ if φ is such that $f_\varphi = f$).

Example 4.4. 1. $f = xy \vee \overline{w}z$ then xy, $\overline{w}z$ are implicants. These are not all the implicants of f.

2. Every term of a DNF representation of f is an implicant.

Definition 4.5. Let f be a Boolean function and C_1, C_2 be two implicants of f. Then C_2 *absorbs* C_1 if $C_1 \leq C_2$

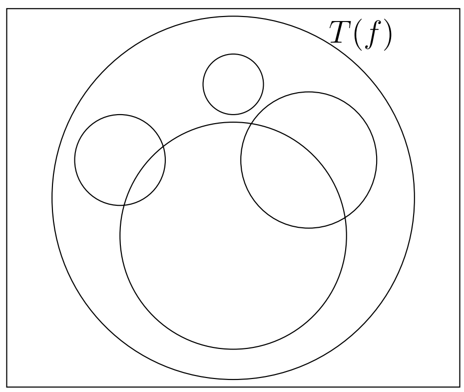
We can represent this as a Venn-Diagram as follows: Here, C_1, C_2 are implicants of f, and



C_1 absorbs C_2.

Definition 4.6. C_1 is a *prime implicant* of f if there is no other implicant of f that absorbs C_1.

So all the prime implicants of f will appear like this



Lemma 4.7. *Let φ be a DNF representing a Boolean function f and C an implicant of f. Then $\varphi \vee C$ represents f.*

Proof. We have $f_c \leq f$ and $f_\varphi = f$ implies that $f_c \vee f_\varphi = f$ so $\varphi \vee c$ represents f. □

Theorem 4.8. *Every Boolean function f can be represented by the disjunction of its prime implicants*

Proof. Let $P_1, \ldots, P_n$ be the prime implicants of f. Let φ be a DNF of f (Which exists by construction). Then

$$\varphi = \bigvee_{k \in \Omega} C_k$$

By inductively applying lemma (4.7), we have

$$\varphi \vee \bigvee_{1 \leq i \leq n} P_i = \bigvee_{k \in \Omega} C_k \vee \bigvee_{1 \leq i \leq n} P_i$$

still represents f. If C_k is not a prime implicant, then it must be absorbed by some prime implicant P_i, therefore we have

$$\bigvee_{k \in \Omega'} B_k \vee \bigvee_{1 \leq i \leq n} P_i$$

represents f where B_k and P_i are all prime implicants. So we are done. $\square$

Definition 4.9. A DNF $\varphi = \bigvee C_k$ is *prime* if its terms, C_k, are all prime implicants of f_φ.

Definition 4.10. A prime DNF $\varphi = \bigvee C_k$ is *irredundant* if removing any prime implicant from φ, yields a Boolean expression which no longer represents f_φ.

Example 4.11. $g = x\overline{y} \vee \overline{x}y \vee x\overline{z}$ is a prime DNF but $y\overline{z}$ is a prime implicant of f_g therefore g is *redundant.*

Counting prime implicants using a Hypercube.

5 Lecture 5: The hypercube, Implicates, Prime Implicates

Definition 5.1. The *hypercube*, Q_n, is the graph on $V = \mathcal{B}^n$ where for $X, Y \in V$, XY forms an edge if and only if X, Y differ in exactly one coordinate. This essentially traces out the hypercube.

Theorem 5.2. *Q_n can be obtained from two copies of Q_{n-1} by joining corresponding vertices with an edge*

Proof. Exercise, most likely to use induction. $\square$

Definition 5.3. A *subcube* of Q_n is a subgraph isomorphic to Q_k for some $k \leq n$

Definition 5.4. Let $A, B \subseteq [n]$ be disjoint. For some $(x_0, \dots, x_n)$ let $x_i = 1$ for all $i \in A$ and $x_j = 0$ for all $j \in B$. This set of points is a subcube of dimension $n - |A| - |B|$ since we have $n - |A| - |B|$ free parameters. We call this an *elementary subcube.*

Theorem 5.5. *Every subcube of Q_n coincides with some elementary subcube.*

Proof. Exercise*, we can use induction on the dimension of the subcube. $\square$

An exercise from the notes states

Proposition 5.6. *There is a one-to-one correspondence between elementary subcubes of Q_n and elementary conjunctions on n variables (not necessarily all of them)*

Example 5.7. We consider the following elementary conjunction: $x\overline{x_3}x_4\overline{x_7}$ on 7 variables. This corresponds to the elementary subcube defined by $A = \{1, 4\}$, $B = \{3, 7\}$.

We now define previously seen things using the hypercube.

Definition 5.8. A term C is an *implicant* of f if its corresponding subcube is contained in $T(f)$.
It is a *prime implicant* if it is maximal with this property, that is there exists no other subcube which is a superset of this one.

Definition 5.9. A *DNF* of f is a covering of $T(f)$ by the corresponding subcubes of its terms.

Remark 5.10. Orthogonal terms correspond to disjoint subcubes. Therefore, a DNF is *orthogonal* if the subcubes corresponding to its terms are all disjoint.

Definition 5.12. Let f be a Boolean function and D an elementary disjunction. We say that D is an *implicate* of f if $f \leq D$.
D is *prime* if no other implicate of f implies it, that is $\nexists\ D' \ : \ f \leq D' \leq D$

6 Lecture 6: Prime Implicates, Restrictions and Essential variables

Lemma 6.1. *A term C is a prime implicant of f if and only if $D := C^d$ is a prime implicate of f^d*

Proof. Suppose $C \leq f$ then we showed as an exercise that this is equivalent to $f^d \leq C^d = D$. Let C_2 be another term. Then $C \leq C_2 \leq f \iff f^d \leq C_2^d \leq C^d = D$ but since C is prime this is equivalent to saying no such C_2 can exist, equivalently no such C_2^d can exist, equivalently D is a prime implicate of f^d. □

Theorem 6.2. *Every Boolean function can be represented by the conjunction of its prime implicates.*

Proof. Let $P_1, \dots, P_n$ be the prime implicants of f^d. Then, using (4.8) we have

$$\begin{aligned} f^d = \bigwedge P_i &\implies (f^d)^d = \left(\bigwedge P_i\right)^d \\ &\implies f = \bigvee P_i^d \\ &\implies f = \bigvee D_i \end{aligned}$$

where $D_i := P_i^d$ which are prime implicates using our previous lemma.
Moreover, to show these are all the prime implicates of the function, take some prime implicate of f call it D. Then D^d is a prime implicant of f^d so $D^d = P_i$ therefore, $D = P_i^d$ so is included in the expression above. □

6.1 Restrictions and Essential Variables

Definition 6.3. For a Boolean function on $\mathcal{B}^n$ and $k \in [n]$ let $f_{|x_k=1}$ be the Boolean function on $\mathcal{B}^{n-1}$ defined by

$$(x_1, \dots, x_n) \mapsto (x_1, \dots, x_{k-1}, 1, x_k, \dots, x_{n-1})$$

We can define $f_{|x_k=0}$ in a similar way. These are called the *restrictions* of f to $x_k = 1$ and $x_k = 0$ respectively.

Example 6.4. Let $f(x, y, z) = xy \vee y\overline{z}$ then $f_{|y=1} = x \vee \overline{z}$ and $f_{|z=1} = xy$

Remark 6.5. If φ is a Boolean expression of f then by replacing each occurrence of x_k by 1 we obtain a Boolean expression for $f_{|x_k=1}$

Theorem 6.6. *Let f be a Boolean function on $\mathcal{B}^n$. Then*

$$f(x_1, \dots, x_n) = x_k f_{|x_k=1} \vee \overline{x_k} f_{|x_k=0}$$

for all $k \in [n]$. This is known as the Shannon Expansion

Proof. We must check whether the LHS equals the RHS for all $X \in \mathcal{B}^n$. Since the RHS relies on x_k it suffices to consider the following two cases:
Suppose $xk = 1$ then

$$\text{LHS} = f(X) = f(x_1, \ldots, x_{k-1}, 1, x_{k+1}, \ldots, x_n) = f_{|x_k=1} = x_k f_{|x_k=1} = x_k f_{|x_k=1} \vee \overline{x_k} f_{|x_k=0}$$

Similar for $x_k = 0$. □

Remark 6.7. By recursively applying the Shannon Expansion to all variables of f, we obtain an orthogonal DNF representation.

Example 6.8. Consider the Boolean function represented by $x \oplus y \oplus z$. Then we have the following

$$\begin{aligned}
x \oplus y \oplus z &= x(1 \oplus y \oplus z) \vee \overline{x}(0 \oplus y \oplus z) \quad \text{Shannon Expansion at } x \\
&= x(\overline{y \oplus z}) \vee \overline{x}(y \oplus z) \quad \text{Simplify} \\
&= y\big((x(\overline{1 \oplus z}) \vee \overline{x}(1 \oplus z)\big) \vee \overline{y}\big(x(\overline{0 \oplus z}) \vee \overline{x}(0 \oplus z)\big) \quad \text{Shannon Expansion at } y \\
&= y\big(x(\overline{\overline{z}}) \vee \overline{x}(\overline{z})\big) \vee \overline{y}\big(x(\overline{z}) \vee \overline{x}(z)\big) \quad \text{Simplify} \\
&= xyz \vee \overline{x}y\overline{z} \vee x\overline{y}\overline{z} \vee \overline{x}\overline{y}z \quad \text{Simplify}
\end{aligned}$$

Definition 6.9. We say variable x_k is *inessential* for f if $f_{|x_k=1} = f_{|x_k=0}$, otherwise x_k is *essential* for f.

Example 6.10. In the expression $\overline{z} \vee \overline{x}y \vee x \vee \overline{y}z$ every expression is inessential since this expression is equivalent to $\mathbf{1}$.

7 Lecture 7: Essential Variables, Prime Implicants via consensus

Theorem 7.1. *The following statements are equivalent*

1. *x_k is inessential for f*
2. *x_k doesn't appear in any prime implicant of f*
3. *f has a DNF in which x_k doesn't appear.*

Proof. (2) $\implies$ (3) is trivial as you can form a DNF from the prime implicants. Also (3) $\implies$ (1) is trivial since $\text{DNF}_{|x_k=1} = \text{DNF}_{|x_k=0}$ as x_k doesn't appear therefore x_k is inessential for f. Let us show (1) $\implies$ (2). Suppose x_k is inessential for f. Let C be any prime implicant of f say

$$C = C_{AB} = \bigwedge_{i \in A} \bigwedge_{j \in B} \overline{x_j}$$

Suppose for a contradiction that x_k appears in C, so WLOG suppose we have $k \in B$ and let $C' = \bigwedge_{i \in A} x_i \bigwedge_{j \in B - \{k\}} \overline{x_j}$ so that

$$C = C'\overline{x_k} \tag{2}$$

We immediately observe that $C \lneq C'$. We claim $C' \leq f$. (Thus we conclude C is not a prime implicant which is our desired contradiction).
Let $Y = (y_1, \ldots, y_n) \in \mathcal{B}^n$ such that $C'(Y) = 1$. We need to show $f(Y) = 1$. Neither f nor C' depends on x_k (by assumption and construction respectively). So we can assume $y_k = 0$. Then by (2)

$$C(Y) = C'(Y) = 1 \implies C(Y) = 1 \implies f(Y) = 1$$

where the final implication makes use of $C \leq f$ (C is a prime implicant of f). Thus we have proved $C' \leq f$ and also derived a contradiction. This proves our theorem. □

7.1 Prime implicants via consensus

Definition 7.2. Given two terms xC and $\overline{x}D$ we call $C \wedge D$ their *consensus* with respect to x.

Remark 7.3. $C \wedge D$ the consensus, can be thought of as the intersection of the projections of xC and $\overline{x}D$ onto the x-plane.

Definition 7.4 (Consensus procedure)**.** Starting from a DNF formula $\varphi = \varphi_0$, repeat one of

1. (Absorbtion) if φ_i has terms $D \leq C$, then *delete* D.
2. (Consensus) if φ_i has terms x_iC, $\overline{x_i}D$, *add* $\vee CD$ to φ_i to obtain φ_{i+1}, unless $CD \leq E$ for some term E of φ_i

This terminates as an algorithm.

Example 7.5. Let $\varphi_0 = x_1\overline{x_2}x_3 \vee \overline{x_1}\overline{x_2}x_4 \vee x_2x_3x_4$. Our consensus procedure is as follows

1. Apply (2) : $\varphi_1 = \varphi_0 \vee \overline{x_2}x_3x_4$
2. Apply (2) : $\varphi_2 = \varphi_1 \vee x_3x_4$
3. Apply (1) (Twice) : $\varphi_4 = x_1\overline{x_2}x_3 \vee \overline{x_1}\overline{x_2}x_4 \vee x_3x_4$ since ($x_3x_4 \geq \overline{x_2}x_3x_4$ and $x_3x_4 \geq x_2x_3x_4$)

Remark 7.6. The consensus procedure doesn't change the function.

8 Lecture 8: Proof of consensus procedure, Monotone functions

Theorem 8.1. *The consensus procedure applied on any DNF, φ, produces the disjunction of all the prime implicants of f.*

Proof Sketch. Let $\varphi = C_1 \vee C_2 \vee \cdots \vee C_k$ be the outcome. Suppose, for a contradiction, $C \leq C_i$ for all i, is a prime implicant of f_φ. Define

$$S := \{\text{term in variables of } \varphi,\ X \mid X \leq C, X \not\leq C_i \ \forall i\}$$

Then $C \in S$ (indeed $C \leq C$ and $C \not\leq C_i$ since C is a prime implicant).
Choose $C^m \in S$ of smallest (should be maximal?) dimension. We observe C^m omits at least

one variable, call it x. (If it didn't, C^m is a point which cannot happen why?)
Then xC^m and $\overline{x}C^m$ are two subcubes and $xC^m, \overline{x}C^m \notin S$ since they have smaller dimension. Then either ($xC^m \nleq C$ or $xC^m \leq C_i$) for some i and ($\overline{x}C^m \nleq C$ or $\overline{x}C^m \leq C_j$ for some j). Clearly $xC^m \leq C^m \leq C$ and $\overline{x}C^m \leq C^m \leq C$.
So, $xC^m \leq C_i$, $\overline{x}C^m \leq C_j$ for some i, j. Knowing $C^m \nleq C_i, C_j$, we have $C^m \leq D_i$, $C^m \leq D_j$ such that $C_i = cD_i$ and $C_j = \overline{x}D_j$.

Thus $C^m \leq D_iD_j$. The consensus on x of C_i, C_j is D_iD_j. Since our consensus algorithm terminated, there exists r such that $D_iD_j \leq C_r$. Therefore, $C^m \leq D_iD_j \leq C_r$. This contradicts the definition of S. □

8.1 Monotone Functions

Definition 8.2. We say $f : \mathcal{B}^n \to \mathcal{B}$ is *positive* in variable x, if $f_{|x=0} \leq f_{|x=1}$. We say f is *negative* in variable x, if $f_{|x=0} \geq f_{|x=1}$.
We say f is *monotone* in variable x, if it is positive or negative in variable x.

Remark 8.3. We have $f(x_1, \ldots, x_n)$ is positive in x_1 if and only if $f(\overline{x_1}, \ldots, x_n)$ is negative in x_1.

Definition 8.4. We say that f is *positive* / *negative* / *monotone* if f is positive / negative / monotone in each variable.

Definition 8.5. Given $X, Y \in \mathcal{B}^n$, we write $X \leq Y$ if $x_i \leq y_i$ for all $i \in [n]$.

Theorem 8.6. *$f : \mathcal{B}^n \to \mathcal{B}$ is positive if and only if $f(X) \leq f(Y)$ for all X, Y satisfying $X \leq Y$.*

Proof. $\Longrightarrow$: Suppose f is positive. Take $X, Y \in \mathcal{B}^n$ such that $X \leq Y$, so $x_i \leq y_i$ for all $i \in [n]$. Then

$$f(x_1, \ldots, x_n) \leq f(y_1, \ldots, x_n) \leq \cdots \leq f(y_1, \ldots, y_n)$$

where each inequality is since $f_{|x_i=0} \leq f_{|x_i=1}$, of course $f_{|x_i=0} \leq f_{|x_i=0}$ and $f_{|x_i=1} \leq f_{|x_i=1}$ hold trivially. Thus $f(X) \leq f(Y)$.
$\Longleftarrow$: Suppose for all $X, Y \in \mathcal{B}^n$ with $X \leq Y$, we have $f(X) \leq f(Y)$. We need f is positive. We have

$$f_{|x_i=0} = f(x_1, \ldots, x_{i-1}, 0, x_{i+1}, \ldots x_n) \leq f(x_1, \ldots, x_{i-1}, 1, x_{i+1}) = f_{|x_i=1}$$

where the inequality uses our assumption as $(x_1, \ldots, x_{i-1}, 0, x_{i+1}, \ldots x_n) \leq (x_1, \ldots, x_{i-1}, 1, x_{i+1})$. This proves f is positive. □

9 Lecture 9: Properties of monotone functions

Definition 9.1. Let $f : \mathcal{B}^n \to \mathcal{B}$ and $X \in T(f)$. We say X is a *minimal true point* if $\nexists Y \in T(f)$ such that $Y \nleq X$.
We say X is a *maximal false point* if $\nexists\, Y \in T(f)$ such that $X \nleq Y$.

Theorem 9.2. *Let f be a positive Boolean function. Then*

1. *$Y \in T(f)$ if and only if $X \leq Y$ for some minimal true point X.*
2. *$Y \in F(F)$ if and only if $Y \leq X$ for some maximal false point X.*

Proof. (1) $\Longrightarrow$ is always true from the definition. $\Longleftarrow$ is true since $f(X) = 1 \implies f(Y) = 1$ by positivity of f. Hence $Y \in T(f)$. The proof of (2) is similar. □

Definition 9.3. Let φ be a DNF. Then

1. φ is positive in x, if $\overline{x}$ doesn't appear in φ.
2. φ is negative in x, if x doesn't appear in φ.

Example 9.4. For $\varphi = x\overline{y}z\overline{w}$, φ is positive in x, z but negative in y, w.

Theorem 9.5. *The following are equivalent:*

1. *f is positive in the variable x.*
2. *$\overline{x}$ doesn't appear in any prime implicant of f (all prime implicants are positive)*
3. *f has a DNF in which $\overline{x}$ doesn't appear.*

Remark 9.6. x is inessential in f if and only if f is positive and negative in x. This can be observed from (7.1).

Proof. (2) $\implies$ (3) is deduced from the prime implicant representation of f.
(1) $\implies$ (2). Let $C_{AB} = \bigwedge_{i\in A} x_i \bigwedge_{j\in B} \overline{x_j}$ be a prime implicant of f. Suppose, for a contradiction, $x_k = x$ for some $k \in B$.
Let

$$C = \bigwedge_{i\in A} x_i \bigwedge_{j\in B-\{k\}} \overline{x_j}$$

Since $C \leq C_{AB}$, we must have $C \not\leq f$, otherwise we contradict C_{AB} being a prime implicant of f. So $\exists\, Y \in \mathcal{B}^n$ such that $C(Y) = 1$ and $f(Y) = 0$. Using $C_{AB} \leq f$ we must also have $C_{AB}(Y) = 0$.

With $C(Y) = 1$ and $C_{AB}(Y) = 0$, it follows $y_k = 1$. Let

$$Y' = Y \oplus (0, 0, \cdots, 1, \cdots, 0)$$

with the 1 at position k. This makes Y' have a 0 at position k. Then $C_{AB}(Y') = 1$ hence $f(Y') = 1$. But $Y \leq Y'$ and $f(Y) \not\leq f(Y')$ which contradicts positivity of f.

(3) $\implies$ (1). Suppose $\varphi = \bigvee_{j=1}^{m} C_j$ is a DNF of f which doesn't contain $\overline{x}$ and $f = f_\varphi$. Then this implies $f_{|x=0} \leq f_{|x=1}$. So f is positive in x. □

Theorem 9.7. *Let φ_1, φ_2 be DNFs and φ_2 be positive. Then*

$$\varphi_1 \leq \varphi_2 \iff \textit{each term of } \varphi_1 \textit{ is absorbed by some term of } \varphi_2$$

Proof. $\Longleftarrow$ is always true, easy. For $\Longrightarrow$, let $C = \bigwedge_{i \in A} x_i \bigwedge_{j \in B} \overline{x_j}$ be a term of φ_1. Let e_A be the characteristic vector of A so

$$e_A = \begin{cases} 1 & \text{if } i \in A \\ 0 & \text{if } i \notin A \end{cases}$$

Then $C(e_A) = 1$, hence $\varphi_1(e_A) = 1$. By assumption, $\varphi_1 \leq \varphi_2$ so $\varphi_2(e_A) = 1$ so there exists a term C' of φ_2 such that $C'(e_A) = 1$. We also know φ_2 is positive (so there are no negations on any literals), therefore $C' = \bigwedge_{i \in F} x_i$. Hence $F \subset A$ and so $C \leq C'$. □

10 Lecture 10: Positivity and Prime Implicant Representation

Theorem 10.1. *Let f be a positive Boolean function. Then the prime implicant representation of f is positive and irredundant.*

Proof. Since f is positive, it is positive in all variables. Then by (9.5), the prime implicants of f contain no negated literal. Thus the prime implicant representation of f is also positive by (9.5).

Let $P_1, \ldots, P_m$ be the prime implicants of f and let

$$\varphi = \bigvee_{i=1}^{m} P_i$$

Suppose, for a contradiction, $f_\varphi = f_\psi$ where ψ is defined as

$$\psi = \bigvee_{i=1}^{m-1} P_i$$

(without loss of generality we choose to omit P_m so we are assuming f is redundant.) Then $f_\varphi = f_\psi = f$ therefore, $f_\varphi \leq f_\psi$. Then (9.5) says φ and ψ are positive. Then using (9.7), we have each term of φ is absorbed by some term of ψ. Specifically, P_m is absorbed by some P_i, $1 \leq i \leq m-1$. This is a contradiction since P_m is a prime implicant and φ, ψ represent the same Boolean function. Thus $f_\varphi \neq f_\psi$ and f is indeed irredundant. □

Remark 10.2.

1. The prime implicant representation for positive functions is the smallest representation in terms of length of literal and disjunctions.
2. $f \equiv 1$ is its own prime implicant representation, by convention 1 is a term. Similarly 0 is a clause.

Theorem 10.3. *Let $\varphi = \bigvee_{k=1}^{m} \left(\bigwedge_{i \in A_k} x_i \bigwedge_{j \in B_k} \overline{x_j} \right)$ be a DNF of a positive Boolean function, f. Then*

$$\psi = \bigvee_{k=1}^{m} \bigwedge_{i \in A_k} x_i$$

also represents f. Moreover, the prime implicants of f are the terms of ψ that are not absorbed by other terms of ψ.

Proof. $f = f_\varphi \leq f_\psi$ is clear. We claim $f_\psi \leq f_\varphi$. Choose some $Y \in \mathcal{B}^n$ such that $\psi(Y) = 1$. Then, we know $\exists k \leq m$ such that $y_i = 1$ for all $i \in A_k$. Hence, $\varphi(Y) = 1$.

Let ψ' be the complete DNF of f. Then $\psi' \leq \psi$ (they are actually equal as shown above). Since ψ is positive by construction, each term of ψ' is absorbed by some prime implicant (term) of ψ. Therefore, $\psi \equiv \psi'$ otherwise we get a contradiction. □

Example 10.4. Is $\varphi = xyzw \vee \overline{x}\,\overline{y}\,\overline{z}\,\overline{w}$ a positive Boolean function?
Answer: Suppose φ is a DNF of a positive Boolean function. Then by (10.3), $\psi = xyzw \vee 1 = 1$ also represents f. But $\varphi(0,1,1,1) = 0$ but $\psi(0,1,1,1) = 1$. Contradiction! So φ does not represent a positive Boolean function.

Example 10.5. $f = x \oplus y \oplus z$ can be represented by $xyz \vee \overline{x}y\overline{z} \vee x\overline{yz} \vee \overline{xy}z$ by the Shannon expansion. Then assuming it is positive tells us $\psi = xyz \vee y \vee x \vee z = x \vee y \vee z$ also represents f. This is a contradiction since these two expressions don't match at $(1,1,0)$.

Theorem 10.6. *Let f be a positive Boolean function and $C_A = \bigwedge_{i \in A} x_i$. Then*

1. *$C_A \leq f \iff e_A \in T(f)$*
2. *C_A is a prime implicant of $f \iff e_A$ is a minimal true point of f*

Proof. (1) $\implies$ Since $C_A(e_A) = 1$ we have $f(e_A) = 1$ so $e_A \in T(f)$.
$\impliedby$ If $e_A \in T(f) \implies \bigwedge_{i \in A} x_i \bigwedge_{j \notin A} \overline{x_j} \leq f$. By positivity, we can omit the negated literals hence

$$\bigwedge_{i \in A} x_i \leq f$$

Therefore $C_A \leq f$.
(2) Given $C_B = \bigwedge_{i \in B} x_i$ such that $C_A \leq C_B$, this is equivalent to $B \subseteq A$ which is equivalent to $e_B \leq e_A$. Therefore we have the following set of equivalences

$$\begin{aligned} C_A \text{ is a prime implicant } &\iff \text{No such } C_B \text{ exists} \iff \text{No such } e_B \in T(f) \\ &\iff e_A \text{ is a minimal true point} \end{aligned}$$

□

11 Lecture 11: Duality, Self-dual functions

Definition 11.1. A Boolean function f is called

1. *dual-minor* if $f \leq f^d$
2. *dual-major* if $f^d \leq f$
3. *self-dual* if $f = f^d$

Example 11.2. Let $f = x_1x_2x_3$ then $f^d = \overline{\overline{x_1x_2x_3}} = x_1 \vee x_2 \vee x_3$ so we have $f \leq f^d$ but not the reverse implication. So f is only dual-minor.

Let $f = x_1 \vee x_2 \vee x_3$ then indeed $f^d \leq f$ so f is only dual-major.

Let $f(x, y, z) = x$. Then $f^d = f$ so f is self-dual.

Remark 11.3. f is dual-minor $\iff$ f^d is dual-major.

The above examples motivate the following theorem

Theorem 11.4. *Suppose x_1 is a prime implicant of $f(x_1, \ldots, x_n)$. Then f is dual-major. Moreover, f is dual-minor if and only if f has no other prime implicants.*

Proof. For the first claim, we have $f = x_1 \vee g(x_1, \ldots, x_n) = x_1 \vee g(x_2, \ldots, x_n)$ we can omit the x_1 in g since any term with x_1 will be absorbed by the term, x_1.
Then $f^d = x_1 \wedge g^d(x_2, \ldots, x_n)$ (by using the dual definition for an expression).
Suppose $f(X) = 0$, then $x_1 = 0$ so $f^d(Y) = 0$. By the contrapositive we have shown $f^d \leq f$ so f is dual-major.

For the second claim, $\impliedby$ Suppose $f(x_1, \ldots, x_n = x_1$, then f is self-dual clearly.
$\implies$: Suppose, for a contradiction, C is another prime implicant i.e. $T(C) \subseteq T(f)$. C must contain a point, Y, with $y_1 = 0$ and $f(Y) = 1$ (otherwise $C \leq x_1$). But $f^d(Y) = 0$ which means $f \not\leq f^d$, but this contradicts f being dual-minor. $\square$

Theorem 11.5 (Restatement of the definitions)**.** *We have*

1. *f is dual-minor* $\iff$ $\forall\ X \in T(f),\ \overline{X} \in F(f)$ $\iff$ $\forall\ X \in \mathcal{B}^n,\ f(X) \wedge f(\overline{X}) = 0$
2. *f is dual-major* $\iff$ $\forall\ X \in F(f),\ \overline{X} \in T(f)$ $\iff$ $\forall\ X \in \mathcal{B}^n,\ f(X) \vee f(\overline{X}) = 1$
3. *f is self-dual* $\iff$ $\forall\ X \in \mathcal{B}^n,\ f(X) = 1$ $\iff$ $f(\overline{X}) = 0$

Example 11.6. Suppose $f : \mathcal{B}^n \to \mathcal{B}$ is self-dual, what is the value of $|T(f)|$. Answer: We pair truth points with false points in a bijection as per the above definition therefore $|T(f)| = \frac{2^n}{2} = 2^{n-1}$.

Suppose $T(f(x, y, z)) = \{(1, 0, 0), (1, 1, 1), (0, 1, 1), (0, 1, 0)\}$. Then there are two opposite points both true which implies f is not self-dual.

12 Lecture 12: Dual-minor and Self-dual functions related

Definition 12.1. f is *maximally dual-minor* if f is dual-minor and for all dual-minor g with $f \leq g$, we infact have $f = g$.

Theorem 12.2. *f is self-dual if and only if f is maximally dual-minor*

Proof. ($\implies$): Suppose f is self-dual and $f \leq g$ for some dual-minor function g. Then $g^d \leq f^d$, $f^d = f$ (self-dual), $f \leq g$ (assumption), $g \leq g^d$ (dual-minor). So we have the list of inequalities

$$g^d \leq f^d = f \leq g \leq g^d$$

Thus $f = g$.
($\impliedby$): We assume $f \leq f^d$ (f dual-minor) and suppose, for contradiction, $f \neq f^d$. Then there exists $Y \in \mathcal{B}^n$ such that $f(Y) = 0$ and $f^d(Y) = 1$.

1. If $y_1 = 1$ then define $g := f \vee (f^d \wedge x_1)$ so that $f \leq g$, we know $f(Y) \neq g(Y)$ so $f \neq g$ and this contradicts f being maximally dual-minor.

2. If $y_1 = 0$ then define $g := f \vee (f \wedge \overline{x_1})$ so that $f \leq g$, we know $f(Y) \neq g(Y)$ so $f \neq g$ and this contradicts f being maximally dual-minor.

For these to be contradictions to the definition of maximally dual-minor, we need to show g is constructed to be dual-minor. We have for the first case

$$g^d = f^d \wedge (f \vee x_1) = f^d f \vee f^d x_1 = f \vee f^d x_1 = g$$

This shows g is self-dual (stronger than dual-minor which is all we need). □

Theorem 12.3. *f is self-dual if and only if f is minimally dual-major*

Proof. Similar proof to previous theorem. □

12.1 Constructing self-dual functions given any arbitrary function

Definition 12.4. The *self-dual extension* of a Boolean function $f(x_1, \ldots, x_n)$ is

$$f^{SD}(x_1, \ldots, x_n, x_{n+1}) := f(x_1, \ldots, x_n)\overline{x_{n+1}} \vee f^d(x_1, \ldots, x_n)x_{n+1}$$

Theorem 12.5. *f^{SD} is self-dual for all Boolean functions, f.*
Moreover, the map $SD : f \mapsto f^{SD}$ is a bijection from

$$\{BFs \text{ on } \mathcal{B}^n\} =: F_n \text{ to } \{SD\ BFs \text{ on } \mathcal{B}^{n+1}\} =: S_{n+1}$$

Proof. We have

$$\begin{aligned}(f^{\text{SD}})^d = (f^d \vee \overline{x_{n+1}}) \wedge (f \vee x_{n+1}) &= (f^d \vee \overline{x_{n+1}}) \wedge f \vee (f^d \vee \overline{x_{n+1}}) \wedge x_{n+1} \\ &= f^d f \vee f\overline{x_{n+1}} \vee f^d x_{n+1} \vee \overline{x_{n+1}} x_{n+1} \\ &= f^d f \vee f^{\text{SD}} \\ &= f^{\text{SD}} \quad \text{as } f^d f \leq f^{\text{SD}} \text{ (absorption)}\end{aligned}$$

So f^{SD} is self-dual. Now for the bijection: We know the cardinalities of f and f^{SD} are equal so it sufficies to show the map is injective.

We know $f_{|x_{n+1}} = f$ so f is injective.
For completeness, surjectivity is shown as: Let $g \in S_{n+1}$. Then $g_{|x_{n+1}=0} \in F_n$

$$\begin{aligned}
g_{|x_{n+1}=0} = g'^{\mathrm{SD}} &= g'\overline{x_{n+1}} \vee g'^{d} x_{n+1} \\
&= g'\overline{x_{n+1}} \vee (g_{|x_{n+1}=0})^{d} x_{n+1} \\
&= g'\overline{x_{n+1}} \vee (g^{d})_{|x_{n+1}=1} x_{n+1} \quad \text{using } f^{d}_{|x=0} = (f^{d})_{|x=1} \\
&= g'\overline{x_{n+1}} \vee g_{|x_{n+1}=1} x_{n+1} \quad \text{using } g^{d} = g \text{ by assumption} \\
&= g_{|x_{n+1}=0}\overline{x_{n+1}} \vee g_{|x_{n+1}=1} x_{n+1} \\
&= g \quad \text{using the Shannon Expansion in reverse}
\end{aligned}$$

□

Theorem 12.6. *If f is dual-minor, then $f^{SD} = f \vee f^{d} x_{n+1}$*

Proof. $f \leq f^{d}$ tells us $f^{\mathrm{SD}} = f\overline{x_{n+1}} \vee f^{d} x_{n+1} = f \vee f^{d} x_{n+1}$ where the last equality is basic, just consider truth values of both sides. □

13 Lecture 13: Positive functions as hypergraphs

Theorem 13.1. *A Boolean function f is positive if and only if f^{d} is positive.*

Proof. ($\Longrightarrow$): f is positive $\iff$ f is positive in all its variables iff $\overline{x_k}$ doesn't appear in any prime implicant, for all variables of f. So the complete DNF of f is positive therefore

$$f = \bigvee_{A \in \mathcal{P}} \bigwedge_{i \in A} x_i$$

where $\mathcal{P}$ represents the prime implicants of f by the indicies of the literals. Then

$$f^{d} = \bigwedge_{A \in \mathcal{P}} \bigvee_{i \in A} x_i$$

which when distributed using the distributive property of Boolean expressions, yields a positive DNF representation for f^{d} thus f^{d} is positive in all its variables and therefore f^{d} is positive.

For ($\Longleftarrow$): We apply the same argument to f^{d} while making use of $(f^{d})^{d} = f$. □

Remark 13.2. We have the following correspondence

$$\{\text{Boolean functions on } \mathcal{B}^{n}\} \stackrel{\text{1-1}}{\longleftrightarrow} \mathcal{P}(\mathcal{P}([n]))$$

$$\{\text{Positive Boolean functions on } \mathcal{B}^{n}\} \quad \stackrel{\text{injective}}{\longrightarrow} \quad \mathcal{P}(\mathcal{P}([n]))$$

The above can be expressed more formally as the correspondence between Boolean functions and hypergraphs.

Definition 13.3. A *hypergraph* $H = (V, E)$ consists of a vertex set V and a hyperedge set $E \subseteq \mathcal{P}(V)$

Definition 13.4. A hypergraph $H = (V, E)$ is called a *clutter* if for all $e, f \in E$ we have $e \subsetneq f$ and $f \subsetneq e$.

Example 13.5. Given a positive Boolean function, f, we associate the corresponding hypergraph as $H_f := ([n], \mathcal{P})$ as defined in the theorem above. Then H_f is a clutter.

Conversely, given a clutter $H = ([n], E)$, the Boolean expression associated to it is

$$\varphi = \bigvee_{A \in E} \bigwedge_{i \in A} x_i$$

which is the complete DNF of some positive Boolean expression. The constraint of being a clutter ensures no hyperedge is contained in another which is equivalent to no implicant being absorbed by another, hence prime.

Definition 13.6. Given a hypergraph $H = (V, E)$ call $U \subset V$ a *transversal* if it intersects each $e \in E$.
It is minimal if no $U' \subsetneq U$ is a transversal.

Theorem 13.7. *Let $f = \bigvee_{A \in \mathcal{P}} \bigwedge_{i \in A} x_i$ and $g = \bigvee_{A \in \mathcal{T}} \bigwedge_{j \in A} x_j$ be complete DNF's of two positive Boolean functions on $\mathcal{B}^n$. (So $([n], \mathcal{P}), ([n], \mathcal{T})$ are clutters). Then the following are equivalent*

1. $g = f^d$
2. *For every partition $[n] = X \coprod Y$ either there exists $A \in \mathcal{P}$ such that $A \subseteq X$ or there exists $b \in \mathcal{T}$ such that $B \subseteq Y$ but not both.*
3. *$\mathcal{T}$ is exactly the* family of minimal transversals

14 Lecture 14: Proof of duality of positive functions and minimal transversals

We begin with the following key lemma into proving (13.7).

Theorem 14.1. *Let $f = \bigvee_{i=1}^{m} \left(\bigwedge_{j \in P_i} x_j \bigwedge_{j \in N_i} \overline{x_j} \right)$ be a Boolean function and $C_{PN} = \bigwedge_{j \in P} x_j \bigwedge_{j \in N} \overline{x_j}$ be an elementary conjunction. Then*

1. *C_{PN} is an implicant of $f^d \iff (P \cap P_i) \neq \emptyset$ or $(N \cap N_i) \neq \emptyset$ for all $i \leq m$*
2. *C_{PN} is a prime implicant of $f^d \iff (i)$ holds and no subset of P, N satisfies (i) i.e for all $P' \subseteq P$, $N' \subsetneq N$ (or other way) there exists $i \in P' \cap P_i = \emptyset$ and $N' \cap N_i = \emptyset$*

Example 14.2 (Algorithm to find implicants of f^d)**.** Let $f = x_1\overline{x_2} \vee \overline{x_1}x_2$. What are the prime implicants of f^d?

We let $C_{PN} = x_1$ ($P = \{1\}, N = \emptyset$). This intersects the first term but not the second, so fails to be an implicant of f^d.
We let $C_{PN} = x_1\overline{x_2}$ ($P = \{1\}, N = \{2\}$). This intersects with the first term but not the second.
We let $C_{PN} = x_1x_2$ ($P = \{1, 2\}, N = \emptyset$). This intersects with both terms hence is an implicant of f^d. In fact it is a prime implicant as one can see.
Same for $C_{PN} = \overline{x_1 x_2}$.

Remark 14.3. $C_{PN} \leq f^d \iff C_{PN}$ agrees with each term of f (any DNF representation) in at least one literal (pure or negated).

Example 14.4 (Algorithm to find prime implicants of f^d)**.** Let $f = x_1\overline{x_2 x_3} \vee \overline{x_1 x_2}x_3$. We want to find the prime implicants of f^d.

We iterate through possibilities of C_{PN} to get $\overline{x_2}$ is a prime implicant of f^d (it is a point). No other prime implicant can have $\overline{x_2}$. So we have x_1x_3 is another. Then $\overline{x_3x_1}$ is another. So

$$f^d = \overline{x_2} \vee x_1x_3 \vee \overline{x_3x_1}$$

is the complete representation of f^d

Remark 14.5. We can apply the algorithm again to f^d to get the prime implicant representation of f.

Proof of theorem. We claim $C_{PN} \leq f^d \iff C_{NP} \leq \overline{f}$. Let $C_{NP} = \bigwedge j \in P\overline{x_j} \bigwedge_{i \in N} x_i$. Then indeed, we have $C_{PN}(X) \leq f^d(X)$ for all $X \in \mathcal{B}^n$. But $f^d(X) = \overline{f(\overline{X})}$. By a change of variables, $Y = \overline{X}$, we have

$$\iff C_{PN}(\overline{Y} \leq \overline{f(Y)} \iff C_{NP}(Y) \leq \overline{f(Y)}$$

for all $Y \in \mathcal{B}^n$ where the second equivalence is by the definition of C_{NP} reversing the pure and negated literals. So $C_{PN} \leq f^d \iff C_{NP} \leq \overline{f}$.

Now we use the claim to prove the original theorem. For ($\implies$) : suppose $C_{PN} \leq f^d \iff C_{NP} \leq \overline{f}$. We know $f \wedge \overline{f} = \mathbf{0}$ and each $C_{P_iN_i} \leq f$ as they are terms of f. Thus we have $C_{P_iN_i} \wedge C_{NP} = \mathbf{0}$ (otherwise $f \wedge \overline{f}$ is true for this specific term conjunction, contradiction). So $P \cap P_i = \emptyset$ or $N \cap N_i = \emptyset$. This completes the forward direction.

For ($\impliedby$): Our condition of $(P \cap P_i) = \emptyset$ or $(N \cap N_i) = \emptyset$ implies C_{NP} clashes with each $C_{P_iN_i}$ so $f \wedge C_{NP} = \mathbf{0}$ hence $C_{NP} \leq \overline{f} \iff C_{PN} \leq f^d$, as required.

The second part (ii): among all those C_{PN} satisfying (i), the prime ones are those with minimal P and N. ☐

Lemma 14.6. $g = f^d \iff g(\overline{X}) = \overline{f(X)}$ *for all* $X \in \mathcal{B}^n$

Proof. $g = f^d \iff g(X) = \overline{f\overline{X})} \iff g(\overline{Y} = \overline{f(Y)}$ for all $Y \in \mathcal{B}^n$ ☐

Proof of minimal transversals theorem. See notes. It is not expected to reproduce this proof and is only for ideas. □

15 Lecture 15: Complexity Measures

Definition 15.1. Let f be a Boolean function and $Y \in \mathcal{B}^n$. A *certificate* of f on Y is a set $S \subseteq [n]$ such that for all points $Z \in \mathcal{B}^n$ with $z_i = y_i$ (for all $i \in S$), we have $f(Z) = f(Y)$.

Definition 15.2. The *certificate complexity*, $C(f, Y)$ of f on Y is the size $|S|$ of a smallest certificate, S, of f on Y.
Then we define the *certificate complexity of f* by

$$C(f) := \max_{Y \in \mathcal{B}^n} C(f, Y)$$

We specifically define the following two, *0-certificate complexity* and *1-certificate complexity* by

$$C_0(f) := \max_{Y \in F(f)} C(f, Y), \quad C_1(f) := \max_{Y \in T(f)} C(f, Y)$$

Lemma 15.3. *To quickly calculate complexities, we have*

$$C(f, Y) = n - \textit{dim(largest subcube containing } Y \textit{ such that } f \textit{ is constant)}$$

Lemma 15.4. *Let S be a* 0*-cerficiate of $Y \in F(f)$ and S' be a* 1*-certificate of $Z \in T(f)$. Then $S \cap S' \neq \emptyset$.*

Proof. Suppose, for a contradiction, $S \cap S' = \emptyset$. Then there exists $W \in \mathcal{B}^n$ such that $w_i = y_i$ for all $i \in S$ and $w_j = z_j$ for all $j \in S'$ (since S, S' are disjoint). Then

$$f(W) = f(Y) = 0, \text{ and } f(W) = f(Z) = 1$$

which is a contradiction. □

Definition 15.5. A K-DNF/K-CNF is a DNF/CNF where each term/clause has at most k literals.

16 Lecture 16: Complexity, Sensitivity, Block-Sensitivity

Lemma 16.1. *f can be written as a K-DNF/K-CNF if and only if $C_1(f) \leq k/C_0(f) \leq k$.*

Proof. For DNFs:
($\implies$): Let φ be a $K - DNF$ of f. Then for all $Y \in T(f)$, there exists a term

$$T = x_{a_1} \cdots x_{a_m} \overline{x_{b_1}} \cdots \overline{x_{b_r}}$$

with $m + r \leq k$. Then we see $S = \{a_1, \ldots, a_m, b_1, \ldots, b_r\}$ is a certificate of f on Y (this would make the term T true), so $C_1(f) \leq k$.

($\Longleftarrow$): Suppose $C_1(f) \leq k$. Given $Y \in T(f)$ and a certificate S_Y on Y, write a term T_Y in the variables in S_Y such that $T_Y(Y) = 1$ (just observe where Y has 0's and 1's). Then our desired K-DNF is

$$\bigvee_{Y \in T(f)} T_Y$$

We can also prove the CNF case in a similar way. $\square$

Definition 16.2. Given $X \in \mathcal{B}^n$ and $S \subseteq [n]$, let $X^S \in \mathcal{B}^n$ be obtained by flipping all coordinates in X in S.

Remark 16.3. S is a certificate on $Y \iff f(Y) = f(Y^B)$ where $B = S^C$ for all $B \subseteq [n] \backslash S$

Definition 16.4. The *sensitivity* $S(f, Y)$ of f on $Y \in \mathcal{B}^n$ is the number of indicies such that $f(Y) \neq f(Y^i)$. Then the sensitivity of f is

$$S(f) := \max_{Y \in \mathcal{B}^n} S(f, Y)$$

Definition 16.5. The *block sensitivity*, $bs(f, Y)$ is the maximum number of disjoint subsets $B_1, \ldots, B_b \subset [n]$ such that $f(Y) \neq f(Y^{B_i})$ for all $i \leq b$. Then the block sensitivity of f is

$$bs(f) = \max_{Y \in \mathcal{B}^n} bs(f, Y)$$

Easily, one can observe that $s(f) \leq bs(f)$.

Theorem 16.6. *We have*

$$bs(f) \leq C(f) \leq S(f) \cdot bs(f)$$

17 Lecture 17: Decision Trees

This definition is best seen using an example.

Definition 17.1. A *decision tree* is a rooted, directed, binary tree such that

1. Every leaf is labelled 0 or 1.

2. Every non-leaf is labelled by a variable $x_1, \ldots, x_n$ and has exactly 2 out-edges labelled 0 and 1.

Remark 17.2. Given a decision tree, we can associate a Boolean function f, by intepretig the leaves labelled 1 as the true points, $T(f)$.

Lemma 17.3. *Given any Boolean function, we can construct a corresponding decision tree.*

Proof. We induct on the dimension of f. For $n = 0$, we take the trivial single leaf 0 or 1 tree. For $n > 0$, let T_0, T_1 be decision trees representing the restrictions $f_{|x_1=0}$ and $f_{|x_1=1}$ respectively. Then, we can construct the decision tree for f using these smaller trees using the inductive hypothesis. $\square$

Given a decision tree, T, and $Y \in \mathcal{B}^n$, let $\text{cost}(T, Y)$ be the length of the path from the root of T to the leaf corresponding to Y.

Definition 17.4. The *decision tree complexity* of f is

$$D(f) := \min_{T \text{ representing } f} \max_{Y \in \mathcal{B}^n} \text{cost}(T, Y)$$

Theorem 17.5. *For all Boolean functions, f, we have*

$$bs(f) \leq D(f)$$

Proof. Given a disjoint set of sensitive blocks $B_1, \ldots, B_b$ of $Y \in \mathcal{B}^n$, we have any decision tree of f must examine at least 1 bit from each B_i. Therefore,

$$D(T) \geq b \implies D(f) \geq \text{bs}(f)$$

□

Theorem 17.6. *For all Boolean functions, f, we have*

$$D(f) \leq C_1(f) \cdot C_0(f) (\leq C(f)^2 \leq bs(f)^4)$$

18 Lecture 18: Functional Completeness Closedness

Definition 18.1. Let F be a set of Boolean functions and define the *set of expressions over F* recursively as follows:

1. Every $f \in F$ is an expression over F, for any number of variables
2. If $c_1, \ldots, c_n$ are expressions over F and $f \in F$ has n variables, then $f(c_1, \ldots, c_n)$ is an expression.

Example 18.2. Every Boolean expression is an expression over $\{\overline{x}, x \wedge y\}$.

Definition 18.3. A set F of Boolean functions is *complete* if every Boolean function can be represented by an expression over F.

Example 18.4. The sets $\{\overline{x}, x \wedge y\}$ and $\{\overline{x}, x \vee y\}$ are complete. Interestingly, $\{\overline{xy}\}$ is also complete.

Theorem 18.5. *Suppose F is complete, and each $f \in F$ can be represented as an expression of some set G. Then G is also complete.*

Proof. Let f be an arbitrary Boolean function. Represent f over F that is

$$f = f_1(f_2(x_1, \ldots), f_3(x_1, \ldots), \ldots, f(x_1, \ldots))$$

Then replace each f_i by expressions in G and we yield an expression for f over G. So every Boolean function can be expressed over G, as required. □

Definition 18.6. The *closure*, denoted $[F]$, of F is the set of functions representable over F. We say F is *closed* if $F = [F]$.

So it follows directly that F is complete $\iff$ $[F] = \{\text{all Boolean functions}\}$.

Theorem 18.7. *For every set of Boolean functions, F, F', we have*

1. $[[F]] = [F]$
2. $F \subseteq F' \implies [F] \subseteq [F']$
3. $[F] \cup [F'] \subseteq [F \cup F']$

Theorem 18.8 (Post's Functional Completeness Theorem)**.** *F is complete $\iff$ F is not a subset of any of the following*

1. *Self-dual functions, S*
2. *Positive functions, P*
3. *Linear functions, L*
4. $T_0 := \{f \mid f(\boldsymbol{0}) = 0\}$
5. $T_1 := \{f \mid f(\boldsymbol{1}) = 1\}$

Lemma 18.9. *T_0 and T_1 is closed.*

Proof. If $f, f_1, \ldots, f_n \in T_0$, then observe

$$f(f_1(\mathbf{0}), \ldots, f_n(\mathbf{0})) = 0$$

so T_0 is closed. □

One can show as an exercise $T_0^d = T_1$.

19 Lecture 19: Sketching the proof of Post's Theorem

Proof. (Proof Sketch) ($\implies$): Suppose $F \subseteq X = \{T_0, T_1, S, P, L\}$. Then (we will show) $[X] = X$ so $[F] \subseteq [X]$. Assuming F is complete, $[F] = F$ hence it follows $[X] = \{\text{all Boolean functions}\}$, contradiction.
($\impliedby$): Suppose $F \not\subseteq$ any such X. We will prove $\{\overline{x}, x \wedge y\} \subseteq F$ implying F must be complete. □

The following series of lemmas help build the theory for proving Post's Theorem.

Lemma 19.1. *We have $[S] = S$, so the set of self-dual functions is closed.*

Proof. Suppose $f, f_1, \ldots, f_m \in S$ and let $F = f(f_1, \ldots, f_m)$. Then

$$F^d = f^d(f_1^d, \ldots, f_m^d) = f(f_1, \ldots, f_m) = F$$

where in the penultimate equality, we make use of $f, f_1, \ldots, f_m$ being self-dual.
It remains to check that $f(x_1', \ldots, x_n') \in S$ for all $f \in S$ and $x_j' = \{x_i$. This follows by setting $y_j = x_j' \in S$ as a Boolean function itself then applying the above. □

Lemma 19.2. *We have $[P] = P$, so the set of positive functions is closed.*

Proof. Suppose $f, f_1, \ldots, f_m \in P$ and we define F as

$$F = f(f_1, \ldots, f_m)$$

Pick some $X \leq Y \in \mathcal{B}^n =:$ domain(F) Then for all $i \leq m$, we have

$$f_i(X) \leq f_i(Y)$$

by the postivity of f_i. More precisely, $f_i(X^{(i)}) \leq f_i(Y^{(i)})$, where $X^{(i)}$ is the subvector of X consisting of coordinates appearing in f_i. Therefore,

$$(f_1(X^{(1)}), \ldots, f_m(X^{(m)})) \leq (f_1(Y^{(1)}), \ldots, f_m(Y^{(m)}))$$

which precisely states $F(X) \leq F(Y)$ hence $F \in P$, as required. We also check $f(x'_1, \ldots, x'_m) \in P$ by setting $y_j = x'_i$ and using the above obtain positivity. □

Lemma 19.3. *Suppose $f(x_1, \ldots, x_n) \notin S$, then by substituting each variable by x or $\overline{x}$, we can obtain a constant function (on 1 variable).*

Proof. Since $f \notin S$, there exists $Y \in \mathcal{B}^n$ such that

$$f^d(Y) = \overline{f(\overline{Y})} \neq f(Y)$$

or, $f(\overline{Y}) \neq \overline{f(Y)}$. So $f(\overline{Y}) = f(Y)$. Now, let

$$\varphi_i(X) = X^{y_i} := \begin{cases} X & \text{if } y_i = 1 \\ \overline{X} & \text{if } y_i = 0 \end{cases}$$

Let $\varphi(x) = f(\varphi_1(x), \ldots, \varphi_n(x))$. We claim φ is constant.

$$\varphi(0) = f(\varphi_1(0), \ldots, \varphi_n(0)) = f(0^{y_1}, \ldots, 0^{y_n}) = f(\overline{y_1}, \ldots, \overline{y_n})$$

where the last equality comes from $0^{y_i} = \overline{y_i}$, $1^{y_i} = y_i$ from the definition above. Therefore, $\varphi(0) = f(\overline{Y}) = f(Y)$.
One can similarly show $\varphi(1) = f(Y)$ too, which shows φ is indeed constant, as required. □

20 Lecture 20: Representations over GF(2)

Note that $x \oplus y = x + y \pmod 2$ and $\mathrm{GF}(2) = \mathbb{F}_2$. We can associate the addition of GF(2) to the operation$\oplus$ and the multiplication to $\wedge$.

Theorem 20.1. *For all Boolean functions $f : \mathcal{B}^n \to \mathcal{B}$, there is a unique $c : \mathcal{P}([n]) \to \mathcal{B}$ such that*

$$f = \bigoplus_{A \in \mathcal{P}([n])} c(A) \bigwedge_{i \in A} x_i \tag{3}$$

Remark 20.2. The number of expressions as in (3) is $2^{\#\mathcal{P}([n])} = 2^{2^n}$, the same as the number of Boolean functions on $\mathcal{B}^n$. Hence, existence of (3) implies uniqueness.

Proof. We induct on the dimension, n. Let $n = 1$, then $f \in \{0, 1, x, \overline{x}\}$. These can all be expressed as in (3). Suppose the result holds for $n > 1$. We make the following claim:

$$f = f_{|x_n=0} \oplus x_n f_{|x_n=0} \oplus x_n f_{|x_n=1}$$

Indeed, when $x_n = 0$, both sides are equal to $f_{|x_n=0}$. And when $x_n = 1$, both sides are equal to $f_{|x_n=1}$ by addition modulo 2.
By our inductive hypothesis, we can express each of $f_{|x_n=0}$ and $f_{|x_n=1}$ by the polynomial in (3). Then using the claim proved above, we can write f as in (3), as required. This completes our induction.

As previously mentioned in our remark, uniqueness follows from the equality in cardinality. □

Example 20.3. Let $f = x_1 \vee x_2$. Then we know f can be represented as

$$f = a \cdot x_1 x_2 \oplus b \cdot x_1 \oplus c \cdot x_2 \oplus d$$

For some $a, b, c, d \in \{0, 1\}$. Solving for these, we get

$$\begin{aligned} x_1 = x_2 = 0 &\implies 0 = d \\ x_1 = 0,\ x_2 = 1 &\implies 1 = c \oplus 0 = c \\ x_1 = 1,\ x_2 = 0 &\implies 1 = b \oplus 0 = b \\ x_1 = x_2 = 1 &\implies 1 = a \oplus 1 \oplus 1 \oplus 0 = a \end{aligned}$$

Therefore, $x_1 \vee x_2 = x_1 x_2 \oplus x_1 \oplus x_2$

Definition 20.4. A Boolean function is *linear* if there exists $c_i \in \{0, 1\}$ such that

$$f = c_0 \oplus c_1 x_1 \oplus c_2 x_2 \oplus \cdots \oplus c_n x_n$$

We denote by L, the set of all linear functions.

Lemma 20.5. *We have $[L] = L$, so the set of linear functions is closed.*

Proof. Combining linear polynomials using the GF(2) representation yields another linear polynomial. So we have closure. □

21 Lecture 21: Completing Proof of Post's Theorem

Lemma 21.1. *If $f \notin L$, then by substituting 0, 1, x or $\overline{x}$ for its variables, we can obtain xy or $\overline{xy}$.*

Example 21.2. Let $f(x, y) = xy \oplus ax \oplus by$ with $a, b \in \{0, 1\}$. Then

$$f(x \oplus b, y \oplus a) = (x \oplus b)(y \oplus a) \oplus a(x \oplus b) \oplus b(y \oplus a) = xy \oplus ab$$

So depending on whether $ab = 0$ or 1, we have the above is equal to xy or $\overline{xy}$.

Lemma 21.3. *Suppose $f \notin P$, Then by substituting its variables with 0, 1 or x, we can obtain $\overline{x}$.*

Proof. This uses an argument seen in graph theory, refer to Lecture 21 notes for the diagram. □

Proof of Post's theorem. ($\Longrightarrow$): Suppose F is complete, so $[F] = \{\text{all Boolean functions}\}$, then immediately, $F \nsubseteq A \in \{T_0, T_1, S, P, L\}$, since each class is closed and none of these classes equals {all Boolean functions}.

($\Longleftarrow$): Suppose $F \nsubseteq A \in \{T_0, T_1, S, P, L\}$. Let $f_0 \notin T_0$, $f_1 \notin T_1$, $f_S \notin S, f_P \notin P, f_L \notin L$ but all of them are in F.
Assume $\{x_1, \ldots, x_n\}$ contains all of the variables across f_0, f_1, f_S, f_P, f_L. We claim

$$\{\overline{x}, xy\} \subseteq [F]$$

which would imply F is complete. We have two cases

1. *Case 1*: If $f_0(\mathbf{1}) = 1$, then $\varphi(x) := f_0(x, x, \ldots, x) \equiv 1$ and $\varphi \in [F]$ since $f_0 \in [F]$. Also $\psi(x) := f_1(\varphi(x), \ldots, \varphi(x)) = f_1(\mathbf{1}) = 0$ and $\psi \in [F]$ since $f_1 \in [F]$. So we have shown $0, 1 \in [F]$. From this, we apply a previous Lemma to conclude $\overline{x} \in [F]$ and then another Lemma to conclude xy or $\overline{xy} \in [F]$ which completes our proof.

2. *Case 2*: If $f_0(\mathbf{1}) = 0$, then $\varphi(\mathbf{1}) = 0$ and $\varphi(\mathbf{0}) = 1$ so $\varphi = \overline{x} \in [F]$. Then by the above argument we are also done.

□

22 Lecture 22: Quadratic Boolean Functions

Corollary 22.1. *If F is closed and $F \neq$ {all Boolean functions}, then F is contained in one of T_0, T_1, S, L, P.*

Theorem 22.2. *Every complete F contains a complete subset of at most 4 elements.*

It should be noted that the 4 is optimal.

Definition 22.3. A *basis* of F is a minimal $F' \subseteq F$ such that $[F'] = [F]$.

Theorem 22.4 (Post)**.** *Every closed class F has a finite basis.*

Therefore, there are countably many closed classes of Boolean functions.

22.1 Quadratic Boolean Functions, Matched Graph

Definition 22.5. A DNF/CNF is *quadratic*, if each of its terms/clauses has at most 2 literals.

Definition 22.6. A Boolean function, f, is *quadratic*, if it has a quadratic DNF. It is called *dually-quadratic* if it has a quadratic CNF.

Observe the naming *dually-quadratic* is because the dual of the quadratic DNF yields a dually-quadratic CNF.

Definition 22.7. Given a quadratic DNF, φ, define its *matched graph*, G_φ, by

$$V := \{x_1, \ldots, x_n, \overline{x_1}, \ldots, \overline{x_n}\} \text{ and}$$
$$E := \{xy \mid x \wedge y \text{ is a term of } \varphi\}$$

Theorem 22.8. *Given a quadratic DNF, φ,*

$$\varphi = 0 \iff \mu(G_\varphi) = \tau(G_\varphi)$$

where μ is the maximum matching size between the pure and negated terms, (which is always n) and τ is the minimal vertex covering of the graph G_φ.

One can use a vertex covering to obtain the solution corresponding to $\varphi = 0$ by assigning uncovered vertices to 1 and covered vertices to 0.

23 Lecture 23: Implication Graph

24 Lecture 24: Proof of Implication Graph

25 Lecture 25: Horn Functions

26 Lecture 26: Relating Horn functions to *union-closed sets conjecture*

27 Lecture 27: Conditional results on *union-closed sets conjecture* using Horn functions

28 Lecture 28: (Non-examinable)

29 Lecture 29: (Non-examinable)

30 Lecture 30: (Non-examinable)